

UDAY A S

Software Engineer II

INTRODUCTION

To build a career in transformative technologies, including Agentic AI, blockchain, robotics, and Brain-Computer Interfaces, among others. My objective is to design state-of-the-art products that push boundaries, while cultivating my expertise and achieving sustained growth on both professional and personal levels.

SKILLS

- Python
- Java
- C/C++
- Haskell
- JavaScript
- React
- NodeJS
- Django
- Artificial Intelligence and Machine Learning
- Retrieval Augmented Generation
- Transformers
- Agentic AI
- Langchain & Langgraph
- MySQL
- AWS
- MongoDB
- Blockchain
- Data Structures and Algorithms
- CI/CD, Docker

EDUCATION

RV College of Engineering

B.E Information Science and Engineering
Aug 2019 - July 2023
CGPA: 8.29

Ananth PU College

11th - 12th
June 2017 - March 2019
Result: 95%

CONTACT

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WORK EXPERIENCE

Software Engineer II

Cisco | August 2023 - Present

- Designed and implemented a new end-to-end approach for enabling service user login in the appliances through tunnels server under tight timelines.
- Significantly reduced the time for VMware tagging by over 80% by redesigning the process as a multithreaded background task.
- Worked extensively on Django, Pexpect, VMware, Celery, and multithreaded programming methodologies, tailored to multienvironment requirements.
- Reduced 10% of workload of operations teams by expanding the platform's capabilities through the addition of multiple features, most notably automated the ability to replace VIPs.
- Authored a crucial program to generate and add certificates to all the prod appliances, reducing manual effort by 99%.
- Engineered a solution to automate prod appliance replacement, ensuring configuration retention and reducing manual workload by 90% in such cases.
- Spearheaded product integration efforts over 4 release cycles, aligning with our sister team's evolving feature set and reducing cross-team dependencies by 80%.
- Implemented service user login to appliance via Kerberos, replacing the legacy method. This change significantly improved the system design, security, and code maintainability.
- Designed and implemented a multithreaded program to attach custom tags to servers deployed in VMware vSphere.
- Conducted talks on topics like "Art of Functional Programming" to a broader audience.

Technical Undergraduate Intern

Cisco | Jan 2023 - June 2023

- Designed and developed a secure public-facing form with custom input validation across multiple fields using Django. Utilized Jira APIs to automatically create a ticket for quick action.
- Additionally, the product is deployed in production using Docker and Apache server.
- Improved dev team productivity by 40% by delivering a long-awaited program to detect and integrate hierarchical XML configuration changes to a Mako XML template.

Software Engineer Intern

HealthBadge | Nov 2021 - Feb 2022

- Designed and developed a robust agent-based simulation of multi-dose vaccination policy.
- Implemented an algorithm to assess the effect of spreading of disease by an agent(s).

PROJECTS

Exoplanet detection (Ongoing)

- A transformer model to search and detect planets outside our solar system, with reliable accuracy, using the data generated by live Transiting Exoplanet Survey Satellite by NASA.
- Leverages the well established "Transit light curve" method for the detection.

Blockchain from Ground-up

- A dynamic peer-to-peer blockchain built from scratch inspired by Bitcoin.
- Features include support for multiple connections, wallet, and transaction pools
- Utilizes the Proof-of-Work consensus system to verify transactions.
- Tools: NodeJS, Web sockets, ExpressJS, Jest

Achievements manager

- A website for storing and retrieving of details related to an achievement made by a student of a particular college.
- Supports multiple views depending on the authorized user i.e. Student, Staff.
- Tools: Python, Django, SQLite, HTML/CSS, Bootstrap

HyperDrive

- A GUI-based Google Drive desktop client app that supports uploading, downloading, deleting, and searching of files.
- Tools: Google Drive APIs, Google OAuth, Java, JavaFX

PageRank

- PageRank works by counting the number of links to a page to determine a rough estimate of how important the website is.
- The algorithm uses the Markov chain / Random surfer model to achieve the same.
- Tools: Python, HTML

Autocomplete

- A CLI program for real-time type-ahead auto-completion of user input inspired by bash.
- Tools: C++, GNU Readline

Minesweeper AI

- A GUI based classic Minesweeper game that can be played both by a human and an AI.
- The AI acts as a knowledge-based agent which uses propositional logic to infer the best move.
- Tools: Python, Pygame

Traffic sign detector

- The machine learning model identifies traffic signs from an image using convolutional neural networks at an accuracy of 96% using the GTSRB dataset.
- Tools: Python, Tensorflow, OpenCV, Keras

Who I am

Someone who is driven by a blend of curiosity and inspiration—think Nolan's storytelling, Naruto's tenacity, and da Vinci's polymathic flair. A voracious reader who thrives at the intersection of philosophy, science, history, and economics. Plus wired to explore complex ideas and turn them into action.
